Near-side azimuthal and pseudo-rapidity correlations of neutral strange baryons and mesons in d+Au and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV

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We present results on the near-side of high-p_T triggered di-hadron correlations of neutral strange baryons (Λ, $\bar{\Lambda}$) and mesons (K_S^0) at intermediate p_T(2-6 GeV/c) in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment at RHIC. The near-side di-hadron correlation in ($\Delta\phi$, $\Delta\eta$) contains two structures, a peak which is narrow in $\Delta\phi$ and $\Delta\eta$ consistent with correlations due to jet fragmentation and correlation in $\Delta\phi$ which is broad in $\Delta\eta$ which is attributed dominantly to flow. The dependence of the conditional yield of the jet-like correlation on the trigger particle momentum, associated particle momentum, and centrality are presented. In addition the particle composition of the jet-like correlation is determined using identified associated particles. Data on identified triggers are qualitatively described by PYTHIA, however the composition of the jet-like correlation is not described by PYTHIA. This indicates that the production mechanism for strangeness in jets is still not understood in p+p collisions.

PACS numbers: 25.75.-qRelativistic heavy-ion collisions 25.75.GzParticle correlations and fluctuations

I. INTRODUCTION

Ultra-relativistic heavy-ion collisions create a unique environment for the investigation of nuclear matter at extreme temperatures and energy densities in the laboratory. Measurements of nuclear modification factors [1–4] have shown that the nuclear medium created is opaque to particles with large transverse momentum (p_T). Moreover, anisotropic elliptic flow measurements demonstrate that the medium exhibits partonic degrees of freedom and has properties close to those expected of a perfect fluid [5, 6].

Studies of jets in heavy ion collisions are possible through single particle measurements [1–4], di-hadron correlations [7–17], and measurements of reconstructed jets [3, 18–21]. Measurements of reconstructed jets are useful for studying higher momenta and have provided the most direct evidence for partonic energy loss in the medium. However, di-hadron correlations provide complementary measurements since they enable studies of intermediate momenta, studies where the jet may be so severely modified that it is no longer recognizable as a jet, and studies where the interplay between jets and the medium is important.

Properties of jets have been studied extensively using di-hadron correlations relative to a trigger particle with large transverse momentum at the Relativistic Heavy Ion Collider (RHIC) [7–14] and the Large Hadron Collider (LHC) [15–17]. Systematic studies of associated particle distributions on the opposite side of the trigger particle revealed significant modification [5, 10, 11]. The associated particle distribution on the near side of the trigger particle, the subject of this paper, is also significantly modified in central Au+Au collisions [5, 7, 8]. In p+p and d+Au collisions, there is a peak narrow in azimuth ($\Delta\phi$) and pseudorapidity ($\Delta\eta$) around the trigger particle, which we refer to as the jet-like correlation. In Cu+Cu and Au+Au collisions this peak is observed to be broader than that in d+Au collisions, although the yields are comparable [14]. This indicates that the jet-like correlation is dominantly produced through fragmentation.

Besides the shape modifications of jet-like correlations at intermediate transverse momentum range, the production mechanism of hadrons seems to differ from simple fragmentation. Baryon production is less suppressed in central Au+Au collisions than that of mesons resulting in baryon to meson ratios enhanced in central Au+Au collisions relative to p+p and d+Au collisions [22, 23]. The measured baryon to meson ratios increase with p_T until reaching their maximum value of ≈ 3 relative to p+p collisions at $p_T \approx 3$ GeV/c in both the strange and non-strange quark sectors. A fall-off of the baryon to meson ratio is observed for $p_T > 3$ GeV/c and both the strange and non-strange baryon to meson ratios in Au+Au collisions approach the values measured in p+p collisions at $p_T \approx 6$ GeV/c.

In this paper, studies of two-particle correlations on the near-side in d+Au, Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV measured by the STAR experiment are presented. Results from two-particle correlations in pseudo-rapidity and azimuth for neutral strange baryons (Λ, $\bar{\Lambda}$) and mesons (K_S^0) at intermediate p_T ($2<p_T<6$ GeV/c) in the different collision systems are compared to unidentified charged hadron correlations. Both identified trigger particles associated with charged particles (K_S^0-h, $\Lambda,\bar{\Lambda}$-h) and unidenfitied charged trigger particles associated with identified strange particles (h-K_S^0, h-$\Lambda,\bar{\Lambda}$) are studied. The near-side jet-like yield is studied as a function of centrality of the collision and transverse momentum of trigger and associated particles to look for possible flavor and baryon/meson. The composition of the jet-like correlation is studied using identified associated particles to investigate possible production mechanisms. Results are compared to expectations from PYTHIA.

II. EXPERIMENTAL SETUP AND PARTICLE RECONSTRUCTION

The results presented in this paper are based on analysis of data from d+Au, Cu+Cu, and Au+Au collisions with $\sqrt{s_{NN}}$ = 200 GeV taken in 2003, 2005, and 2004,respectively. Reordered discussion but this paragraph is still very similar to the other paperThe MB trigger for Au+Au collisions was obtained using a ZDC coincidence, a signal in both BBCs and a minimum charged particle multiplicity in an array of scintillator slats arranged in a barrel, the Central Trigger Barrel (CTB), to reject non-hadronic interactions. An additional online trigger for central collisions was used for Au+Au collisions. This trigger was based on the energy deposited in the ZDCs in combination with the multiplicity in the CTB. The central trigger sampled the most central 12% of the total hadronic cross section. For Cu+Cu collisions, the MB trigger was based on the combined signals from the Beam-Beam Counters (BBC) at forward rapidity ($3.3 < |\eta| < 5.0$) and a coincidence between the ZDCs. The d+Au events were selected using a minimally biased (MB) trigger requiring at least one beam-rapidity neutron in the Zero Degree Calorimeter (ZDC), located at 18 m from the nominal interaction point in the Au beam direction, accepting 95$\pm$3% of the hadronic cross section [24].

These sentences are the same but I'm not sure there's another way to order them. In order to achieve a more uniform detector acceptance, only those events with the primary vertex position along the longitudinal beam direction (z) within 30 cm of the center of the STAR detector were used for the analysis. For the d+Au collisions this was expanded to $|z| < 50$ cm. The number of events after the vertex cut in individual data samples is summarized in table II.

STAR [25] is a large acceptance, multi-purpose spectrometer consisting of several detectors inside a large solenoidal magnet with a uniform magnetic field of 0.5 T applied along the beam line. This analysis is based exclusively on charged particle tracks detected and reconstructed in the Time Projection Chamber (TPC) [26]. The TPC has a pseudo-rapidity acceptance $|\eta| < 1.5$ and full azimuthal coverage. The TPC has 45 pad rows in the radial direction allowing up to 45 independent spatial and energy loss (dE/dx) measurements along each charged particle track. The momentum resolution is determined to be $\Delta p/p \sim 0.0078 + 0.0098 \cdot p_T$ (GeV/c) [26].

We identify weakly decaying neutral strange (V^0) particles Λ, $\bar{\Lambda}$ and K^0_S by topological reconstruction of their decay vertices from their charged hadron daughters measured in the TPC as described in [27]:

$$\begin{aligned}
\Lambda &\rightarrow p + \pi^-, BR = (63.9 \pm 0.5)\% \\
\bar{\Lambda} &\rightarrow \bar{p} + \pi^+, BR = (63.9 \pm 0.5)\% \\
K^0_S &\rightarrow \pi^+ + \pi^-, BR = (68.95 \pm 0.14)\%
\end{aligned} \tag{1}$$

where BR denotes the branching ratio. The STAR V^0 finding reconstruction software pairs oppositely charged particle tracks into V^0 candidates. Cuts are optimized for each data set and yield signal to background ratio of at $\geq$ 15:1. For the analyses presented here, no significant difference was observed between results with Λ and $\bar{\Lambda}$ trigger and associated particles so the correlations were added to increase statistics.

TABLE I: Number of events after cuts (see text) in the data samples analyzed.

System	Centrality	No. of events [10^6]
d+Au	0-95%	3
Cu+Cu	0-60%	38
Au+Au	0-80%	28
Au+Au	0-12%	17

III. METHOD

A. Correlation technique

The analysis in this paper follows the method in [14]. Unidentified hadron and V^0 daughter tracks used in this analysis were required to have at least 15 fit points in the TPC. Unidentified hadron tracks were required to have a DCA to the primary vertex of less than 1 cm and a pseudorapidity $|\eta| < 1.0$. Reconstructed Λ and K^0_S particles were required to be within $|\eta| < 1.0$. A high-p_T trigger particle was selected and the raw distribution of associated tracks relative to that trigger in pseudorapidity ($\Delta\eta$) and azimuth ($\Delta\phi$) is formed. This distribution, $d^2N_{\mathrm{raw}}/d\Delta\phi\, d\Delta\eta$, is normalized by the number of trigger particles, N_{trigger}, and corrected for the efficiency and acceptance of associated tracks ε:

$$\frac{d^2N}{d\Delta\phi\, d\Delta\eta}(\Delta\phi,\Delta\eta) = \frac{1}{N_{\mathrm{trigger}}}\frac{d^2N_{\mathrm{raw}}}{d\Delta\phi d\Delta\eta} \frac{1}{\varepsilon_{\mathrm{assoc}}(\phi,\eta)}\frac{1}{\varepsilon_{\mathrm{pair}}(\Delta\phi,\Delta\eta)}. \tag{2}$$

The efficiency correction $\varepsilon_{\mathrm{assoc}}(\phi,\eta)$ is a correction for the TPC single particle reconstruction efficiency and $\varepsilon_{\mathrm{pair}}(\Delta\phi,\Delta\eta)$ is a correction for the finite TPC track-pair acceptance in $\Delta\phi$ and $\Delta\eta$. Since the correlations are normalized per trigger, the efficiency correction is only applied for the associated particle. The data presented in this paper are averaged between positive and negative $\Delta\phi$ and $\Delta\eta$ regions and are reflected about $\Delta\phi$ =0 and $\Delta\eta$ =0 in the plots.

B. Single charged track efficiency correction

Since the correlations are normalized by the number of trigger particles, no correction for the efficiency of the trigger particle is necessary. In the correlations, each track pair is corrected for the efficiency for reconstructing the associated particle. For identified trigger particles, the efficiency correction is the correction for unidentified charged hadrons, identical to the efficiency correction applied in [14]. The single particle reconstruction efficiency in the TPC is determined by simulating the detector response to a particle and embedding these signals into a real event. The efficiency for detecting a single track as a function of p_T, η, and centrality is determined from the number of simulated particles which were successfully reconstructed. The single track efficiency is approximately constant for $p_T > 2$ GeV/c and ranges from around 75% for central Au+Au events to around 85% for peripheral Cu+Cu events. The efficiency for reconstructing a track in d+Au is 89%. The reconstruction efficiency for Λ, $\bar{\Lambda}$, and K^0_S reconstruction is determined in a similar way, but forcing the simulated particle to decay through the channel in equation II and then correcting for the branching ratio. The efficiency for Λ, $\bar{\Lambda}$, and K^0_S ranges from 8% to 15%, increasing with momentum and decreasing with system size. The systematic uncertainty on the efficiency correction, 5%, is strongly correlated across centralities, p_T bins, and particle type for each data set but not between data sets.

C. Pair acceptance correction

With the restriction that each track falls within $|\eta| < 1.0$, there is a limited acceptance for track pairs. For $\Delta\eta \approx 0$, the geometric acceptance of the TPC for track pairs is $\approx$ 100%, however, near $\Delta\eta \approx 2$ the acceptance is close to 0%. In azimuth the acceptance is limited by the 12 TPC sector boundaries, leading to dips in the acceptance of track pairs in azimuth. To correct for the geometric acceptance, a mixed event analysis was done using trigger particles from one event combined with associated particles from another event, as done in [7]. The position of the event vertices were required to be within 2 cm of each other along the beam axis and have the same charged particle multiplicity within 10 particles. Each event with a trigger particle was mixed with ten other events.

D. Track merging correction

The track merging effect in unidentified hadron (h) correlations discussed in [14] is also present for V^0-h and h-V^0 correlations. This effect leads to a dip at small $\Delta\phi$ and $\Delta\eta$ due to overlap between the trigger and associated particles. When one of the particles is a V^0, this overlap is between one of the V^0 daughter particles and the unidentified hadron. The size of the dip due to track merging depends strongly on the relative momenta of the particle pair. This effect is larger when the overlapping particles' momenta are closer. For V^0-h correlations, the typical associated particle momentum is approximately 1.5 GeV/c. Since the K^0_S decay is symmetric, this means

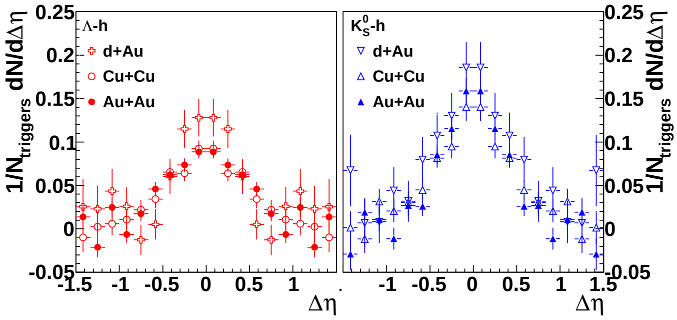

FIG. 1: $\frac{dN_J}{d\Delta\eta}$ for Λ, $\bar{\Lambda}$-h (left) and K^0_S-h (right) correlations for minimum bias d+Au collisions, 0-20% Cu+Cu collisions, and 40-80% Au+Au correlations. The data have been reflected about $\Delta\eta = 0$.

that the track merging effect is greatest for K^0_S-h correlations with a trigger K^0_S momentum of approximately 3 GeV/c. In a Λ decay, the proton daughter carries more of the Λ momentum than the pion daughter. Therefore this effect is larger for Λ trigger particles with lower momenta. Since track merging affects both signal and background particles and the signal sits on top of a large combinatorial background, the effect is larger for collisions with a higher charged track multiplicity. Since the dip in V^0-h and h-V^0 correlations is the result of a V^0 daughter merged with an unidentified hadron, the dip is wider in $\Delta\phi$ and $\Delta\eta$ than in unidentified hadron correlations. When the track merging dip is present, it is corrected by fitting a Gaussian to the peak, excluding the region impacted by track merging, and using the fit to extract the yield.

E. Yield extraction

Extraction of the yield in this paper follows the method and the notation in [7, 14]. The jet-like correlation is narrow in both $\Delta\phi$ and $\Delta\eta$ and is contained within the kinematic cuts in $p_T^{\rm trigger}$ and $p_T^{\rm associated}$ used in this analysis. The di-hadron correlation from equation 2 is projected onto the $\Delta\eta$ axis:

$$\left.\frac{dN}{d\Delta\eta}\right|_{a,b} \equiv \int_a^b d\Delta\phi \frac{d^2N}{d\Delta\phi d\Delta\eta}. \quad (3)$$

All other correlations, including those from v_2, $V_{3\Delta}$, and higher order flow harmonics, are assumed to be independent of $\Delta\eta$ within the η acceptance of the analysis, consistent with [7, 28–30]. This means that the jet-like yield can be determined by:

$$\frac{dN_J}{d\Delta\eta}(\Delta\eta) = \left.\frac{dN}{d\Delta\eta}\right|_{-0.78,0.78} - b_{\Delta\eta} \quad (4)$$

where $b_{\Delta\eta}$ is a constant offset determined by fitting a constant background $b_{\Delta\eta}$ plus a Gaussian to $\frac{dN_J}{d\Delta\eta}(\Delta\eta)$.

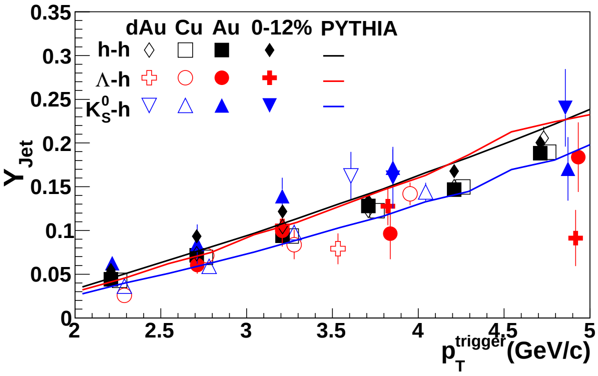

FIG. 2: The jet-like yield as a function of $p_T^{\rm trigger}$ for h-h [14], K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations in d+Au, Cu+Cu, and Au+Au collsions at $\sqrt{s_{NN}}$ = 200 GeV. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. The data are compared to PYTHIA [31] calculations using the Perugia 2011 tune [32].

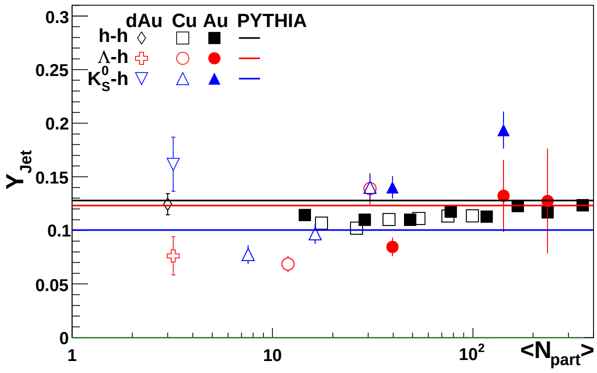

FIG. 3: $N_{\rm part}$ dependence of the jet-like yield of h-h, K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations in d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. The data from Au+Au collisions are shown in two centrality bins, 40-80% and 0-12%. The data are compared to PYTHIA [31] calculations using the Perugia 2011 tune [32].

Sample correlations are given in Figure 1. When the dip due to track merging is negligible, the yield determined from fit is discarded to avoid any assumptions about the shape of the peak and instead we integrate equation 4 over $\Delta\eta$ using bin counting to determine the jet-like yield $Y_J^{\Delta\eta}$:

$$Y_J^{\Delta\eta} = \int_{-0.78}^{0.78} d\Delta\eta \, \frac{dN_J}{d\Delta\eta}(\Delta\eta) \, . \tag{5}$$

When it is not possible, the yield from the Gaussian fit is used, excluding the region affected by track merging, and the statistical errors from the fit exceed any error from the assumption of a Gaussian shape.

IV. RESULTS

A. Correlations with identified strange trigger particles

The jet-like yield as a function of $p_T^{\rm trigger}$ is shown in figure 2 for h-h [14], K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. Due to residual track merging effects, fits are used for $\Lambda,\bar{\Lambda}$-h correlations in Cu+Cu collisions with $2.0 < p_T^{\rm trigger} < 3.0$ GeV/c, in 0-12% Au+Au collisions for $3.0 < p_T^{\rm trigger} < 4.5$ GeV/c, and in 40-80% Au+Au collisions for $2.0 < p_T^{\rm trigger} < 4.5$ GeV/c and for K_S^0-h correlations in 0-12% Au+Au collisions for 3.0 $< p_T^{\rm trigger} <$ 3.5 GeV/c and 40-80% Au+Au collisions for $2.0 < p_T^{\rm trigger} < 4.5$ GeV/c. As observed for h-h correlations in [14], there is no significant difference in the yields between the collision systems. This includes no significant difference between results from Au+Au collisions in 40-80% and 0-12% central collisions, although there is a hint that the yield may be lower in central collisions. The jet-like yield from K_S^0-h correlations is systematically above the jet-like yield from h-h correlations and the jet-like yield from $\Lambda,\bar{\Lambda}$-h is systematically below the jet-like yield from h-h correlations, particularly in Au+Au collisions. Data are compared to PYTHIA calculations from the Perugia 2011 [32] tune. PYTHIA describes the trend of the jet-like yield well for all particle species. However, the trigger particle type ordering in PYTHIA is opposite that observed in the data, with the $\Lambda,\bar{\Lambda}$-h jet-like yieldcomparable to the h-h jet-like yield, which is higher than K_S^0-h jet-like yield. Similar mass ordering is observed in the Perugia 2010 tune.

The jet-like yield as a function of $N_{\rm part}$ is shown in figure 3 for h-h [14], K_S^0-h, and $\Lambda,\bar{\Lambda}$-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. All yields are determined using bin counting. While there is no centrality dependence in the jet-like yield of h-h correlations, there is an increase in the yield with centrality for K_S^0-h correlations and a hint that there may be a similar increase in $\Lambda,\bar{\Lambda}$-h correlations. This could be caused by many different factors. If strange jets experience more energy loss than light flavored jets, the jet-like yield associated with a strange trigger particle could increase even as the jet-like yield stays the same for light particles. If lower energy strange jets are suppressed more than lower energy light jets, the surviving spectrum of jets could be skewed towards higher energy jets and the corresponding jet-like yield would be higher. An increased yield could also come from a modification of fragmentation functions inside the medium, either from an enhancement in the production of lower energy particles in particles containing a strange jet or from an enhancement in the production of higher energy strange particles in light jets. Jana please review the preceeding logic. Jana I also removed the fits to the different particle species since I don't think it'd be very meaningful These

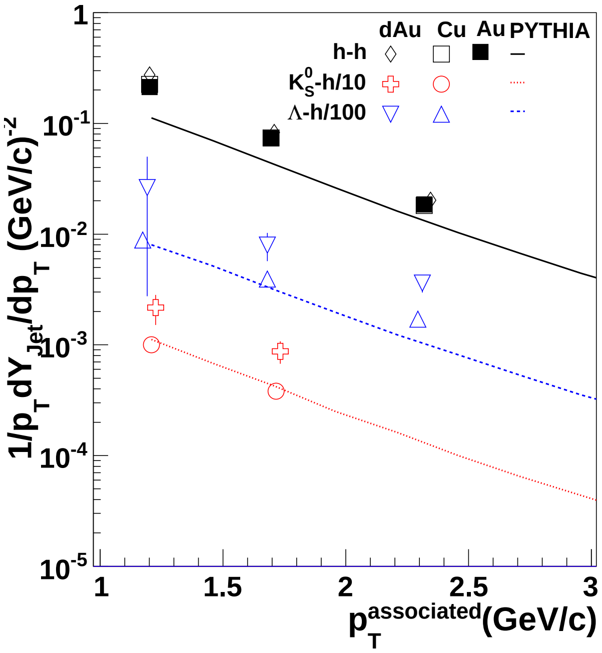

FIG. 4: The jet-like yield as a function of $p_T^{\text{associated}}$ for h-h [14], K^0_S-h, and $\Lambda, \bar{\Lambda}$-h correlations in d+Au, Cu+Cu, and Au+Au collsions at $\sqrt{s_{NN}}$ = 200 GeV. The data are compared to PYTHIA [31] calculations using the Perugia 2011 tune [32].

data are compared to PYTHIA [31] calculations from the Perugia 2011 [32] tune. PYTHIA gets the trigger particle type ordering wrong, even for d+Au collisions, indicating that the production mechanism for strange particles in PYTHIA may need to be refined.

The jet-like yield as a function of $p_T^{\text{associated}}$ is shown in figure 3 for h-h [14], K^0_S-h, and $\Lambda, \bar{\Lambda}$-h correlations for d+Au, Cu+Cu, and Au+Au collisions at $\sqrt{s_{NN}}$ = 200 GeV. All yields are determined using bin counting. The $\Lambda, \bar{\Lambda}$-h and K^0_S-h correlations are only shown for d+Au and Cu+Cu collisions since residual track merging made measurements in Au+Au collisions difficult. These data are compared to PYTHIA [31] calculations from the Perugia 2011 [32] tune. PYTHIA describes the Cu+Cu data better than the d+Au data. Jana, I removed the fits and the PID'd Au+Au points. The Au+Au points all looked kind of iffy - I wondered whether we could really extract a yield. They either had large errors or the background was maybe not flat. I've left all of the plots in the analysis note for now. Overall I think that this is an issue we should discuss.

For all data with identified strange trigger particles, PYTHIA is unable to describe the particle type ordering. The data indicate that there may be limitations in the models for strangeness production in jets, highlighting the need for both better models and more precise data.

B. Charged-V^0 correlations

It is possible to study the composition of the jet-like correlation using identified associated particles. The statistics of the d+Au data were limited and the Au+Au data set was limited by the presence of residual track merging. Therefore it was only possible to extract data from Cu+Cu collisions. Since the jet-like correlation in h-h correlations has been shown to be consistent across collision systems [7, 14], the composition of the jet-like correlation is expected to be the same for all collision systems. These data are shown in figure 5 and compared to data from inclusive spectra in p+p collisions from STAR [33] and ALICE [34] and from PYTHIA [31] calculations from the Perugia 2010 and 2011 [32] tunes. The Cu+Cu data are consistent with the inclusive particle ratios from p+p, which is expected if the dominant production mechanism for strange particles at high momenta is mainly fragmentation. For each PYTHIA tune, the inclusive ratio and the ratio in the jet-like correlation are comparable and lower than that observed in the data. The Perugia tunes, which improved the description of inclusive strangeness production at LHC energies, are further from the observed $\frac{\Lambda+\bar{\Lambda}}{2K^0_S}$ than Tune A. While these tunes are better at describing the yields, none of the PYTHIA tunes available are able to describe strangeness production completely. This may indicate a fundamental problem with strangeness production in PYTHIA.

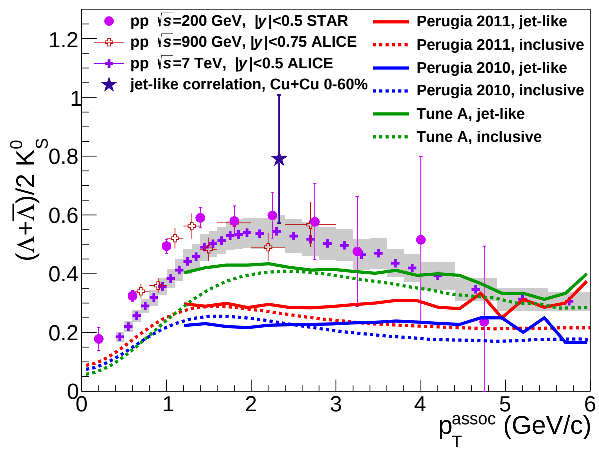

FIG. 5: Λ/K^0_S ratio measured in inclusive p_T distributions, near-side jet-like and ridge-like correlation peaks in Au+Au collisions together with this ratio obtained from inclusive p_T spectra in p+p collisions.

V. CONCLUSIONS

We observe a trend for mass ordering of the yield of identified strange trigger particles in d+Au, Cu+Cu, and Au+Au collisions. This mass ordering is not the same ordering observed for the jet-like yield in PYTHIA. The $\frac{\Lambda+\bar{\Lambda}}{2K^0_S}$ ratio for the jet-like correlation is comparable to that observed in p+p collisions, providing further evidence that this correlation is dominantly produced by fragmentation. In PYTHIA, this ratio is comparable to the inclusive ratio. PYTHIA is unable to describe either the inclusive ratios or the ratio in the jet-like correlation. This indicates that there may still be some problems with the production mechanism for strange particles in PYTHIA.

These studies provide motivation for future studies of strangeness production in jets. Larger data sets and data from collisions at higher energies could provide more robust tests of the strangeness production mechanism in p+p collisions. Studies in p+p would be essential in order to search for modifications of strangeness production in jets in heavy ion collisions.

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